

Sakila Database Mini Capstone Project

Project Overview

This Mini Capstone Project provides students with an opportunity to apply data analytics and SQL skills to a real-world business scenario using the Sakila database. Students will perform exploratory and analytical queries to generate actionable insights that support business decision-making.

Project Objective

The objective of this project is to conduct a comprehensive business analysis of the Sakila database by extracting and analyzing transactional data, identifying key performance trends, generating insights using SQL queries, and providing data-driven recommendations.

Part 1: Business Analysis Questions

1. Determine the total rental revenue generated by each store, including the earliest and latest payment dates.
2. Analyze the month-by-month revenue breakdown for each store.
3. Compare total rental revenue between stores and calculate percentage contribution.
4. Evaluate rentals per month and average payment per category.
5. Rank stores by revenue, rentals, and customer spending.
6. Identify highest revenue-generating film categories.
7. Determine top revenue-generating customers.
8. Analyze monthly rental trends by category.
9. Evaluate staff performance in rentals and revenue.
10. Provide strategic recommendations for revenue growth.

Part 2: SQL Query Development

1. Total Revenue and Reporting Period

Write an SQL query to calculate the **total revenue generated by each store**, including:

- Store ID
- Total revenue

- Earliest payment date
- Latest payment date

Use the payment table and connect payments to stores through the appropriate relationships.

2. Monthly Revenue Breakdown

Write a SQL query to show the **monthly revenue breakdown for each store**, including:

- Store ID
- Revenue month
- Total monthly revenue

Sort the results by store and month.

3. Store Revenue Comparison

Write a SQL query that compares **each store's total revenue to the company total**, showing:

- Store ID
 - Store revenue
 - Total company revenue
 - Percentage of total revenue contributed by each store
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4. Transactions per Month by Category

Write a SQL query to determine the **number of rentals per month** and the **average payment amount per rental category**.

Return:

- Rental month
- Category name
- Number of rentals
- Average payment amount

Use the category, film_category, and rental tables.

5. Store Performance Ranking

Write a SQL query to rank stores based on:

- Total rental revenue
- Total number of rentals
- Average customer payment

Use a ranking function to assign store rankings.

6. Top Revenue Categories per Store

Write a SQL query to identify the **highest revenue-generating film categories for each store**, showing:

- Store ID
- Category name
- Total revenue
- Revenue rank within each store

Use a window function to rank categories.

7. Top Customers by Store

Write a SQL query to rank customers within each store by:

- Total amount spent
- Number of rentals

Return:

- Store ID
- Customer name
- Total spent
- Rental count

- Customer rank
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8. Monthly Rental Trends by Category

Write a SQL query to analyze **monthly rental trends by film category**, showing:

- Month
- Category name
- Total rentals

Then calculate the **month-over-month change in rentals** for each category.

9. Staff Performance Analysis

Write a SQL query to measure staff performance by showing:

- Staff ID
- Staff name
- Total rentals processed
- Total payment revenue processed
- Average revenue per rental

Rank staff members by total revenue.

10. Revenue Growth Opportunity Analysis

Write a SQL query to identify **film categories with declining monthly revenue**, showing:

- Category name
- Current month revenue
- Previous month revenue
- Revenue difference

Sort by the largest revenue decline.

Use this output to determine where management should focus efforts next quarter.

Part 3: Store Revenue and Performance Analysis Report

Project Overview

This project analyzes business performance using the Sakila sample database. The analysis focuses on store revenue, customer behavior, rental trends, staff performance, and film category profitability.

The goal is to transform raw transactional data into actionable business insights that can support strategic decision-making for revenue growth and operational improvement.

Database Description

Database Used

- Sakila Sample Database

Sakila is a fictional DVD rental database commonly used for SQL training and analytics practice. It contains information about:

- Customers
- Stores
- Rentals
- Payments
- Films and categories
- Staff

Business Objective

The primary objectives of this analysis are to:

- Evaluate total and monthly revenue performance by store
- Identify high-performing and under-performing stores
- Analyze rental trends over time
- Determine top revenue-generating film categories
- Measure customer value and spending behavior
- Assess staff performance based on transactions and revenue
- Identify declining revenue categories

- Provide data-driven recommendations for improving next quarter's performance
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Key Performance Indicators (KPIs)

Key KPIs Analyzed

The following key performance indicators (KPIs) were measured:

- Total revenue by store
 - Monthly revenue trends
 - Store contribution percentage to total revenue
 - Number of rentals by category
 - Average payment per rental
 - Top revenue-generating film categories
 - Customer lifetime value (total spend and rental frequency)
 - Staff revenue performance
 - Month-over-month revenue change
 - Declining category revenue trends
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Analytical Approach

The analysis was performed using SQL with the following techniques:

- JOIN operations across multiple tables
 - Aggregation functions (SUM, COUNT, AVG)
 - Date-based grouping and formatting
 - Common Table Expressions (CTEs)
 - Window functions (RANK, LAG)
 - Subqueries for comparative analysis
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Key Insights Summary

- Revenue is concentrated in a small number of high-performing categories.
- Certain stores contribute disproportionately to total company revenue.
- Customer spending varies significantly, with a small group of high-value customers driving a large portion of revenue.
- Some film categories show consistent decline and require strategic intervention.
- Staff performance differs significantly, indicating opportunities for training and optimization.
- Seasonal patterns exist in rental behavior and revenue generation.

Tools & Technologies

- MySQL / MySQL Workbench
 - Sakila Database
 - Word Processing Software
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Strategic Recommendations

1. Focus on declining categories by introducing promotions and refreshing inventory.
 2. Invest in high-performing categories to maximize revenue growth.
 3. Strengthen customer retention through loyalty programs for high-value customers.
 4. Improve underperforming stores by adjusting inventory and local marketing strategies.
 5. Enhance staff performance using training and performance-based incentives.
 6. Align operations with seasonal trends to optimize staffing and inventory planning.
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